Lecture Scribe CSE 400

L20.pdf
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You are generating a Lecture Scribe for CSE 400: Fundamentals of Probability in Computing.
This lecture scribe is intended to serve as exam-ready reference material.

This task follows One-Shot Prompting. A single example lecture scribe is provided below only to demonstrate structure.
You must follow the same structure when generating the lecture scribe for the provided lecture slides.

Requirements

Generate a lecture scribe that:

Uses only the provided lecture slides and context
Is faithful to the lecture content
Is suitable for reading-based / closed-notes exam preparation
Includes definitions, assumptions, results, derivations, and worked examples only if they appear in the lecture
Maintains formal academic tone
Follows logical flow of the lecture
Uses clear headings and structured organization
Does not introduce new examples or explanations
Does not simplify or creatively reinterpret the lecture
Does not include external knowledge

Writing

CSE400 – Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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Date: March 19, 2025

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1. Overview

This lecture introduces the statistical representation of multiple random variables and examines how linear transformations affect their expectations and distributions. The lecture also extends Gaussian distributions from lower dimensions to multivariate settings and discusses covariance-based dependencies.

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2. Lecture Outline

The lecture covers the following topics:

- Statistical Representation of Random Vectors
- Mean Vector, Correlation, and Covariance Matrices
- Linear Transformations
- Mapping Properties under (Y = AX + b)
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of Zero Correlation and Matrix Factorization

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3. Statistical Representation of Random Vectors

3.1 Random Vector

A set of multiple random variables can be represented as a **random vector**:

[
X =
\begin{bmatrix}
X_1 \
X_2 \
\vdots \
X_N

\end{bmatrix}
]

Where:

- (X_1, X_2, ..., X_N) are random variables
- (N) represents the dimensionality of the system

This representation allows modeling multiple dependent variables simultaneously.

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4. Case Study: Smart City Traffic + Weather System

The lecture introduces a case study of designing an intelligent system to predict traffic congestion in a city using both transportation and climate variables.

The system uses multiple variables such as:

- Traffic congestion
- Rainfall
- Temperature
- Pollution
- Speed

These variables are represented using an **N-dimensional random vector**, where:

[
N = 5
]

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5. Mean Vector

The **Mean Vector** represents the expected values of each random variable:

[
\mu =
\begin{bmatrix}
E[X_1] \
E[X_2] \
\vdots \
E[X_N]

$$
\end{bmatrix}
$$

Interpretation:

- Represents typical system behavior
- Describes average traffic and climate conditions
- Used as a baseline for probabilistic analysis

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6. Covariance Matrix

6.1 Definition

The **Covariance Matrix** captures relationships between random variables:

$$
\Sigma =
\begin{bmatrix}
Cov(X_1,X_1) & Cov(X_1,X_2) & \cdots \\
Cov(X_2,X_1) & Cov(X_2,X_2) & \cdots \\
\vdots & \vdots & \ddots
\end{bmatrix}
$$

6.2 Interpretation

The covariance matrix represents dependencies among variables. In the smart city case:

- More rain ⇒ more congestion
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

Thus, the covariance matrix acts as a:

"Dependency Map of the City"

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7. Joint Gaussian Distribution

The lecture assumes the system follows a **Joint Gaussian Distribution**.

Assumption:

The random vector follows:

$$
X \sim N(\mu, \Sigma)
$$

This implies:

- The system follows structured probabilistic behavior
- Mean vector represents typical conditions
- Covariance matrix represents dependencies

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8. Linear Transformations

8.1 Linear Transformation Model

A new variable can be defined as:

$$
Y = AX + b
$$

Where:

- (X) is the original random vector
- (A) is a transformation matrix
- (b) is a constant vector
- (Y) is the transformed random vector

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9. Case Study Transformation: Congestion Index

The city defines a **Congestion Index** as a linear transformation of multiple variables.

This is represented as:

[
Y = AX + b
]

This transformation combines:

- Traffic variables
- Climate variables
- Environmental indicators

into a single congestion metric.

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10. Inference from Linear Transformation

Inference #1

Since the transformation is linear:

- Dependencies between variables are preserved
- Mean and covariance transform systematically

Inference #2

If rainfall increases suddenly:

Effects observed:

- Speed decreases
- Traffic increases
- Pollution increases

This demonstrates how linear transformations propagate changes across correlated variables.

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11. Multivariate Gaussian Distribution

The lecture extends Gaussian distributions from:

- One dimension

- Two dimensions
- N dimensions

General form:

$$
X \sim N(\mu, \Sigma)
$$

Where:

- μ is mean vector
- Σ is covariance matrix

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12. Properties of Multivariate Gaussian

Key properties discussed:

- Fully defined by mean and covariance
- Linear transformation preserves Gaussianity
- Correlation structure captured by covariance matrix
- Joint behavior determined by dependencies

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13. Special Cases

The lecture discusses special covariance scenarios:

13.1 Zero Correlation

If:

$$
Cov(X_i, X_j) = 0
$$

Then:

- Variables are uncorrelated
- Dependencies disappear

13.2 Matrix Factorization

Covariance matrix factorization provides:

- Structural interpretation
- Simplified computation
- Dimensional reduction insight

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14. Summary

This lecture introduced:

- Random vectors
- Mean vectors
- Covariance matrices
- Multivariate Gaussian distributions
- Linear transformations
- Effects of dependencies in probabilistic systems

The Smart City Traffic + Weather case study demonstrated how multiple correlated random variables can be modeled and transformed using linear algebra and probability theory.

End of Lecture 20 — Exam-Oriented Scribe

give it to me in latex code

Here is the **LaTeX code** for your Lecture 20 Scribe following the **same one-shot structure** and **exam-ready format**.

You can directly paste this into Overleaf.

LaTeX

[redacted]